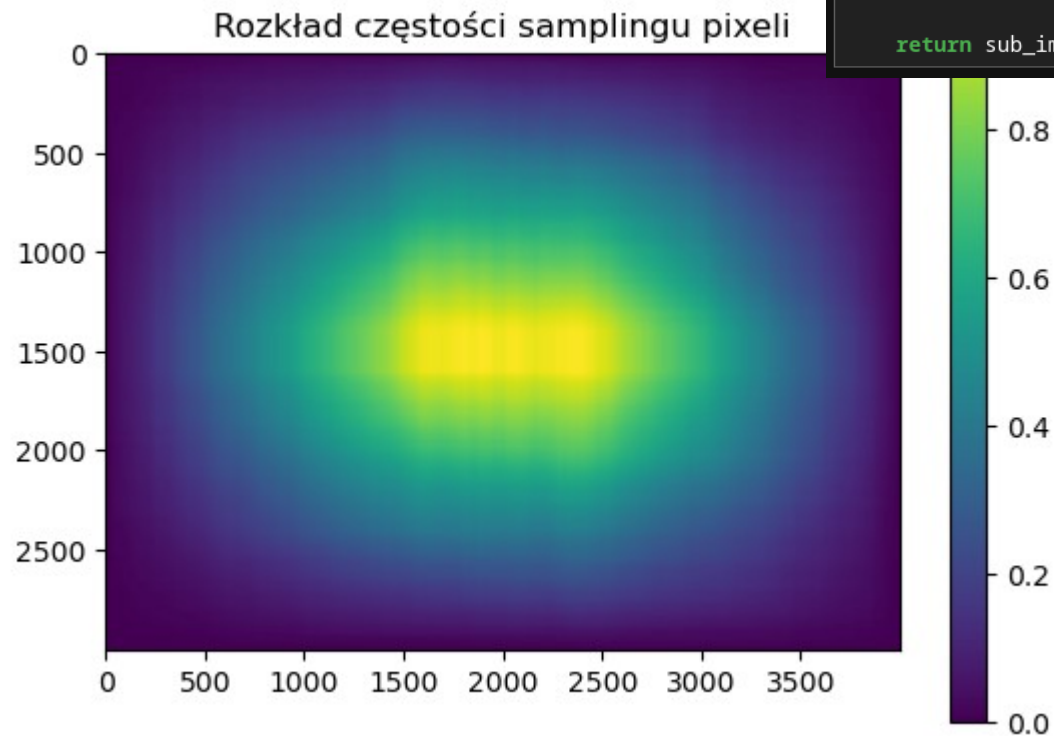


Klasy = {
1: "dynia",
2: "migdal",
3: "goji",
4: "zurawina",
5: "ziemny"
}



130 zdjęć: 4k×3k px jpg

Procedura resamplingu:
256 razy wybieram losowe zdjęcie, a
w nim losowy kwadrat o boku
1600px. Przed zapisaniem zostaje
przeskalowany do 256×256 px jpg



```
def load_img(path):  
    return np.asarray(Image.open(path))
```

```
IMG_SHAPE = (3000, 4000)  
SHIFT_I, SHIFT_J = 1600, 1600  
SHIFT_RANGES = [*map(lambda x, y: x - y, IMG_SHAPE, (SHIFT_I, SHIFT_J))]
```

```
def make_random_shift_img(path):  
    """ loads a random constant size rectangle from given image """  
    img = load_img(path)  
    # upper left corner of img  
    i, j = map(lambda x: random.randint(0, x), SHIFT_RANGES)  
    # get sub image  
    sub_img = img[i:i+SHIFT_I, j:j+SHIFT_J, :]  
  
    return sub_img
```

```
def bootstrap(save_to, files, resize: bool|int=False, filetype: str="jpg"):  
    """ Select random rectangles from original images and downscale  
    save_to: directory where to save images  
    files: list of file paths  
    resize: if not False, resize to square with specified size  
    filetype: png or jpg  
    """  
  
    if filetype not in ("jpg", "png"):  
        raise Exception(f"Not supported type: {filetype}. Pick 'png' or 'jpg'.")  
  
    for i in range(256):  
        file = random.choice(files)  
        img = Image.fromarray(make_random_shift_img(file))  
  
        if resize:  
            img.resize((resize, resize)).save(f"{save_to}/{i:0>3}.{filetype}")  
        else:  
            img.save(f"{pic_dir}/{i:0>3}.{filetype}")  
  
bootstrap(  
    save_to="pics/sub_pics",  
    files=files.proper,  
    resize=256,  
    filetype="jpg"  
)
```

```
NUM_Classes = 5 + 1
NUM_epochs = 30
BATCH_size = 8
```

```
Epoch 10/30 | train loss: 0.1582 | val loss: 0.2442
```

```
EARLY STOPPING: brak poprawy val loss
```

```
Test loss (best-val model): 0.2495
```

```
===== METRYKI =====
```

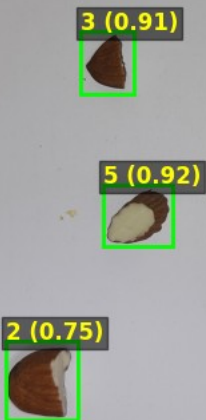
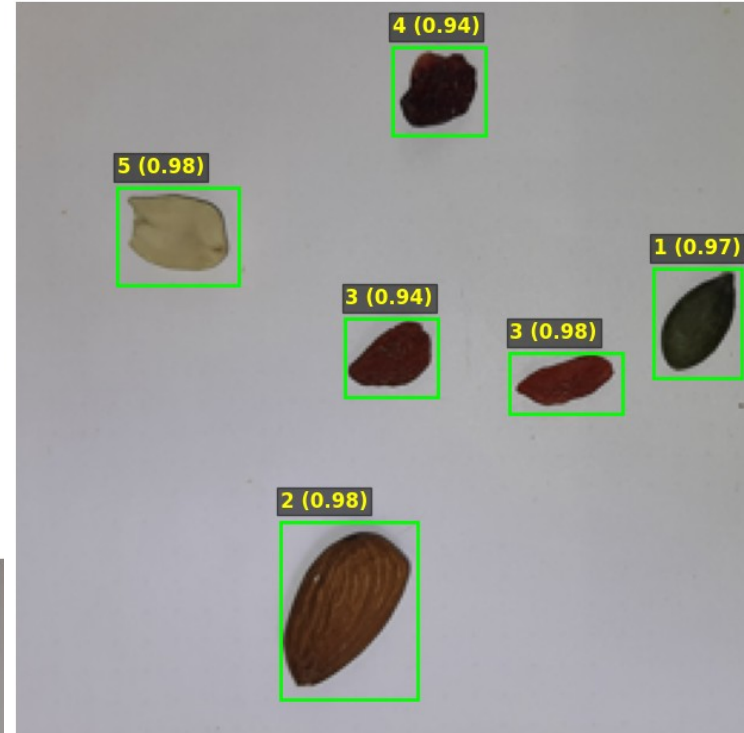
```
Precision: 0.931
```

```
Recall:    0.969
```

```
=====
```







Klasy = {
 1: "dynia",
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 }

